

Programming Fundamentals

Question 1: What is the output of the following code snippet?

```
int x = 7, y = 2;  
cout << x / y;
```

- A) 3.5
 - B) 3
 - C) 3.0
 - D) 4
-

Question 2: Which manipulator forces fixed-point notation and controls decimal places when combined with setprecision?

- A) showpoint
 - B) setw
 - C) fixed
 - D) left
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Question 3: What happens when the following code executes?

```
int age;  
string name;  
cin >> age;  
getline(cin, name);
```

- A) Both inputs work correctly
 - B) The getline() function reads an empty string
 - C) The program crashes
 - D) The age variable receives a string value
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Question 4: Which logical operator would you use to check if a score is between 0 and 100 (inclusive)?

- A) ||
 - B) !
 - C) &&
 - D) &|
-

Question 5: What is the size in bytes of the bool data type on most systems?

- A) 0
 - B) 1
 - C) 2
 - D) 4
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Section B: Short Answer Questions (10 × 2 = 20 marks)

Instructions: Answer any TEN of the following questions. Each question carries 2 marks. Write brief but complete answers.

Question 6: Explain the difference between integer division and floating-point division in C++ with an example of each.

Question 7: What is the purpose of the cin.ignore() function? Provide a scenario where it is necessary to use it.

Question 8: List any four relational operators available in C++ and describe what each one does.

Question 9: What is the difference between the prefix (++i) and postfix (i++) increment operators? Provide an example showing their different behavior.

Question 10: Explain what hand tracing is and why it is a useful debugging technique for programmers.

Question 11: What is a flag in programming? Give an example of a situation where you would use a flag variable.

Question 12: Describe the difference between the #include <iostream> and #include "myheader.h" directives.

Question 13: What is the purpose of the sizeof() operator? Write a code statement that displays the size of a double variable on screen.

Question 14: Explain what short-circuit evaluation means in the context of logical operators (&& and ||). Provide an example.

Question 15: What is the difference between using endl and "\n" for creating new lines in C++ output?

Question 16: Write a complete C++ program that:

- Declares three integer variables
 - Prompts the user to enter values for these variables
 - Calculates and displays their average as a double value
 - Uses appropriate type casting to ensure correct division
 - Formats the output to show exactly 2 decimal places
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Question 17: Write a C++ program that implements a grade classification system. The program should:

- Prompt the user to enter a numeric score (0-100)
- Validate that the input is within the valid range
- Display the corresponding letter grade using the following scale:
 - 90-100: A
 - 80-89: B
 - 70-79: C
 - 60-69: D
 - Below 60: F
- Use if/else if statements to determine the grade
- Display an appropriate error message for invalid input

Question 18: Write a C++ program that creates a simple calculator menu. The program should:

- Display a menu with options: 1) Add, 2) Subtract, 3) Multiply, 4) Divide, 5) Exit
 - Read the user's choice using an integer variable
 - For options 1-4, prompt the user for two numbers and perform the operation
 - Use a switch statement to handle the menu choices
 - Continue showing the menu until the user chooses to exit
 - Handle division by zero appropriately
-

Question 19: Write a C++ program that demonstrates the use of mathematical functions from the cmath library. The program should:

- Prompt the user to enter two coordinates (x1, y1) and (x2, y2)
 - Calculate and display the distance between these two points using the distance formula
 - Calculate and display the power of the first number raised to the second number
 - Calculate and display the square root of the sum of the coordinates
 - Round a floating-point number entered by the user to the nearest integer
 - Include appropriate comments explaining each calculation
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Question 20: Write a complete C++ program that demonstrates the use of the string class. The program should:

- Prompt the user to enter their full name (including spaces)
 - Read the entire line using getline()
 - Display the length of the name
 - Display the first character and the last character of the name
 - Convert the entire name to uppercase (assume you can access characters by index)
 - Create a greeting message by concatenating "Hello, " with the user's name
 - Display the final greeting
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Question 21: Write a C++ program that calculates the final grade for a student. The program should:

- Declare named constants for the weight percentages: Homework (30%), Midterm (30%), Final Exam (40%)
- Prompt the user to enter scores for homework, midterm, and final exam
- Validate that each score is between 0 and 100
- Calculate the weighted average using the constants
- Determine the letter grade using if/else if statements
- Display the final numeric average (formatted with 2 decimal places) and the letter grade
- Use meaningful variable names and include appropriate comments